

SMART INDIA HACKATHON 2026



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HACKATHON
2026

- **Problem Statement ID** – SIH26017
- **Problem Statement Title** – Predictive Analytics System for Early Detection of Land Acquisition Delays
- **Theme** – Smart Automation
- **PS Category** - Software
- **Sponsoring Organization** – Ministry of Rural Development
- **Department** – Department of Land Resources (DoLR)
- **Team ID** – 137891
- **Team Name** – Npointer





Proposed Solution & Approach

1

Data Collection & Integration

Gather data from government records, satellite imagery and project documents

2

Early Warning System

Use AI/ML models to predict potential delays and highlight at-risk projects

3

Spatial Analysis

GIS-based mapping to visualize project status, impacted areas and conflict zones

4

Risk Assessment

Identify key risk factors (legal, social, environmental) and assess severity

5

Decision Support

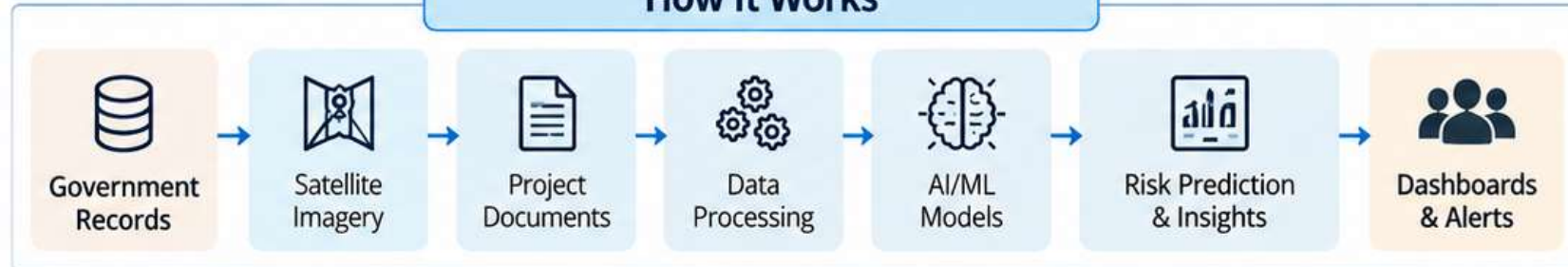
Provide actionable insights and recommendations to officials

6

Public Access & Offline Alerts

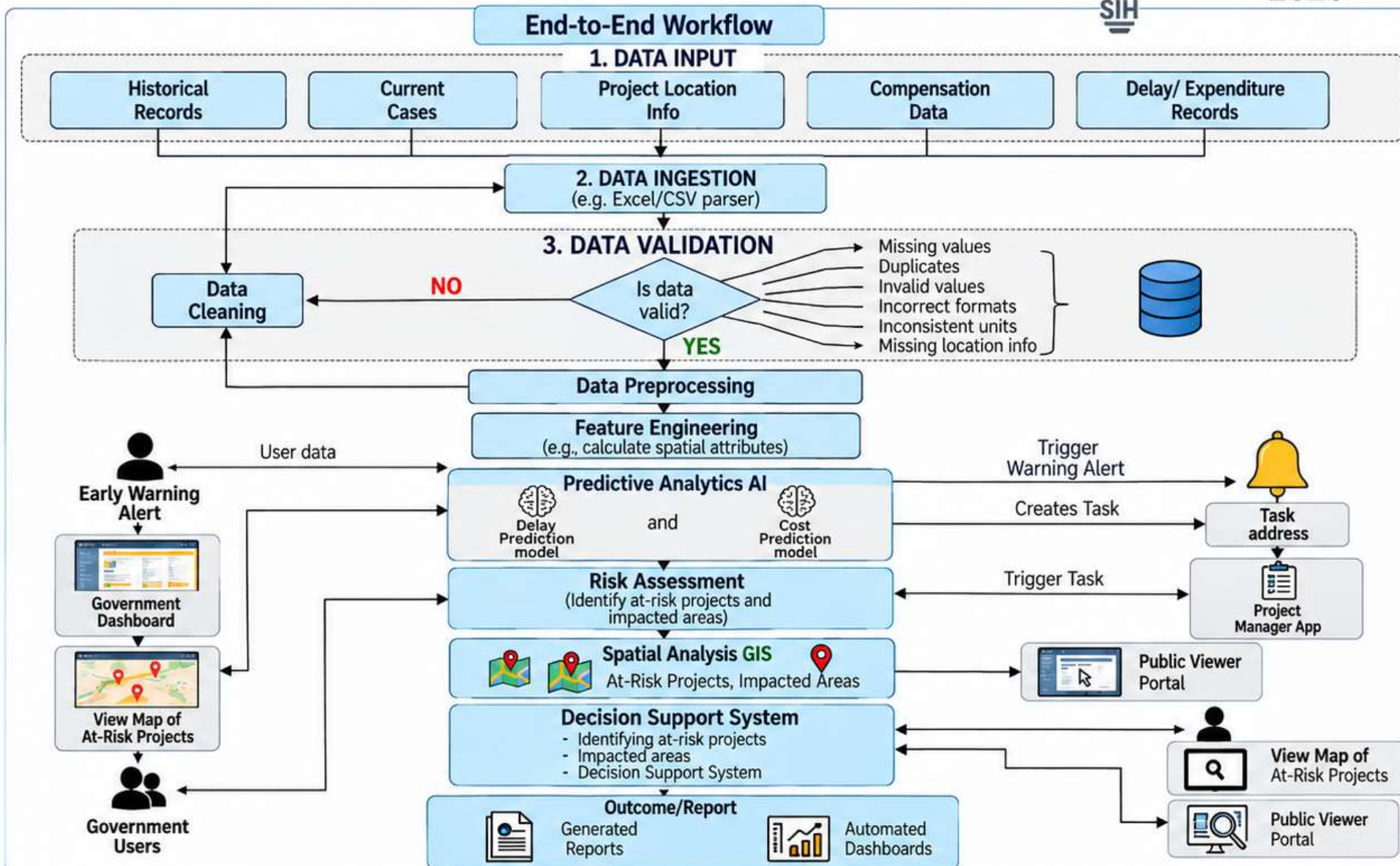
Dashboards for government users and public viewer for transparency

How It Works



Key Features / Modules

**GitHub**<https://github.com/omlokhande27/SIH26017.git>



Feasibility of the Idea

Technical Feasibility

- Uses existing technologies (Python, GIS, ML, Web)
- Open-source tools
- Scalable architecture

Social Feasibility

- Benefits citizens, farmers and communities
- Increases transparency
- Builds trust in the process

Legal & Policy Feasibility

- Aligns with LARR Act 2013
- Supports government transparency goals
- Uses official data sources

Environmental Feasibility

- Helps in planned land use
- Reduces environmental conflicts
- Supports sustainable development

Operational Feasibility

- Works with available government data
- Minimal manual effort
- Automated monitoring

Economic Feasibility

- Cost-effective
- Uses open-source technologies
- Reduces delay-related financial losses

LandGuard AI Feasible

- Scalable
- Practical
- Impactful

Challenges Faced & Strategies to Overcome

Challenges



Data availability & quality issues

- Incomplete or outdated land records
- Different data formats



Integration of diverse datasets

- Different formats and standards



Accurate prediction of delays

- Complex and multi-factor dependencies



Scalability and real-time monitoring

- Large volume of data
- Need for real-time updates



User adoption and accessibility

- Varied user groups (government officials, public)

Strategies to Overcome



- Use multiple data sources (government records, satellite imagery)
- Data cleaning and validation pipeline
- Standardize data formats



- Use open-source GIS tools (PostGIS, GeoJSON)
- Build data integration modules
- Adopt common data standards (OGC)



- Use ML models (classification/regression)
- Include multiple features (legal, social, environmental, historical data)
- Continuous model improvement



- Cloud-based deployment
- Modular and scalable architecture
- Automated data pipelines



- Simple and intuitive dashboard
- Role-based access
- Offline alerts and mobile-friendly interface

BENEFITS



Early Risk Detection

Identify potential land acquisition delays at an early stage



Location-Specific Mapping

Visualize at-risk projects geographically with GIS



Safer Execution Planning

Helps authorities plan better and allocate resources efficiently



Data-Driven Decisions

Provides reliable insights using real government data



Accessible to Officials & Public

Dashboards and maps ensure transparency and easy access to information



IMPACTS



Community Safety

Reduces disputes and ensures fair compensation



Disaster Response

Enables faster action in disaster-prone or critical areas



Economic Loss

Minimizes project delays and related financial losses



Infrastructure

Supports timely development of public infrastructure



Environmental Balance

Helps in better land-use planning and reduces ecological impact

- **National Institute of Rural Development (NIRD)** – <https://nirdpr.org.in>
Reports and studies on land acquisition and rural development.
- **Ministry of Rural Development (MoRD)** – <https://rural.nic.in>
Policies and guidelines related to land acquisition and rehabilitation.
- **Department of Land Resources (DoLR)** – <https://dolr.gov.in>
Official data, rules and notifications on land acquisition.
- **Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013** – <https://legislative.gov.in>
Legal framework for land acquisition in India.
- **Open Government Data (OGD) Platform** – <https://data.gov.in>
Government datasets for land, census, and socio-economic indicators.
- **Bhuvan – ISRO's Geoportal** – <https://bhuvan.nrsc.gov.in>
Satellite imagery and geospatial data.
- **CAG Reports on Land Acquisition** – <https://cag.gov.in>
Audit reports highlighting key issues and delays.
- **GIS and Remote Sensing Resources** – <https://qgis.org> | <https://grass.osgeo.org>
Open-source tools for spatial analysis.
- **Research Papers (IEEE/ScienceDirect)** – <https://ieeexplore.ieee.org>
<https://www.sciencedirect.com>
Studies on predictive analytics, machine learning in governance and land use.
- **Government Portals for Real-time Data** – <https://india.gov.in> | <https://pmgsy.nic.in> | <https://mordashboard.nic.in>
Project status, rural infrastructure and monitoring dashboards.
- **OpenStreetMap** – <https://www.openstreetmap.org>
Base map data for visualization and mapping.

Platform Screenshots

